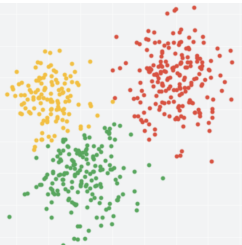


Agglomerative Clustering

Procheta Sen



Agglomerative (bottom-up) methods: main idea

- ▶ The individual objects are successively agglomerated into higher-level clusters.
- ▶ The main variation among the different methods is in the choice of objective function used to decide the merging of the clusters.

Agglomerative (bottom-up) methods: main idea

Input: dataset $\mathcal{D}$

1. **Initialise:** place every object in $\mathcal{D}$ in its own cluster
2. **Repeat:**
 - 2.1 Find *closest* pair of clusters i and j
 - 2.2 Merge clusters i and j
3. **Until** termination criterion
4. **Return** current clustering or hierarchy or clusterings

Measure of proximity between clusters: single-linkage

Let P and Q be two clusters. Assume we have a distance function $d(\cdot, \cdot)$ for objects.

Best (single) linkage: the distance between P and Q is the minimum distance between a pair of objects one of which is in P and the other in Q .

$$\text{dist}(P, Q) = \min_{\bar{X} \in P, \bar{Y} \in Q} d(\bar{X}, \bar{Y})$$

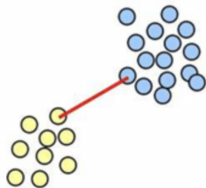
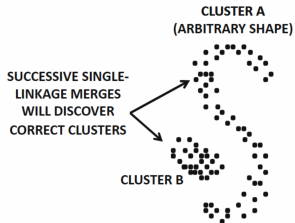
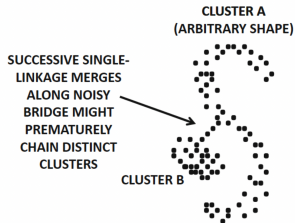


Figure: Single Linkage Demonstration

Single Linkage Clustering



(a) Good case with no noise



(b) Bad case with noise

Figure:

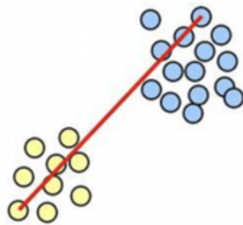
Measure of proximity between clusters: complete-linkage

Let P and Q be two clusters. Assume we have a distance function $d(\cdot, \cdot)$ for objects.

Worst (complete) linkage: the distance between P and Q is the maximum distance between a pair of objects one of which is in P and the other in Q .

$$\text{dist}(P, Q) = \max_{\bar{X} \in P, \bar{Y} \in Q} d(\bar{X}, \bar{Y})$$

Implicitly attempts to minimize the maximum diameter of a cluster (i.e. the largest distance between any pair of points in the cluster).



Measure of proximity between clusters: group-average linkage

Let P and Q be two clusters with p and q objects respectively. Assume we have a distance function $d(\cdot, \cdot)$ for objects.

Group-average linkage: the distance between P and Q is the average distance between all pairs of objects one of which is in P and the other in Q .

$$\text{dist}(P, Q) = \frac{1}{p \cdot q} \sum_{\bar{X} \in P, \bar{Y} \in Q} d(\bar{X}, \bar{Y})$$

